Educational Data Mining

Slide 1: Opening (30–40 seconds)
Assalamu Alaikum and good morning, respected teachers.

This is **Sultan Mahamud Opu**, and my teammates are Najma Akter Lopa and Sourav Ray.

Today, we are going to present our topic titled **"Educational Data Mining."**

Our main focus is to understand what educational data mining is, how it works, and why **Educational Data Mining** is becoming so important in modern education systems.

Slide 2: What is Data Mining (1 minute)

Let's start with **Data Mining**.

Data Mining is the process of discovering **hidden patterns, correlations, and useful information** from large datasets using **machine learning, statistics, and database systems**.

In today's digital world, around **2.5 quintillion bytes of data are generated every day**. Because of this, every organizations are **data-rich but informationly poor**.

The main goal of data mining is to **bridge this gap** by converting raw, unstructured data into **meaningful and actionable information**.

This graph illustrates the rapid growth of data from **2010 to 2025**.

In **2010**, only **2 zettabytes** of data were generated.

In **2013**, this amount became more than triple around **9 zettabytes**.

The data volume continued to increase in **2016, 2019, and 2020**, and by the end of **2025**, it had become **181 zettabytes**.

which clearly highlights the importance of data mining.

Slide 3: What is Educational Data Mining (EDM) (1 minute)

Now, moving to **Educational Data Mining**, or EDM.

EDM is a specialized field of data mining that focuses on **educational data** such as student grades, attendance records, and assignment submissions etc.

The main purpose of EDM is to **understand student learning behavior**, **analyze academic performance**, and support **data-driven educational decision-making**.

The Ultimate goal of EDM is to **improve learning outcomes** and make education more effective and personalized.

Slide 4: KDD Process and EDM Process (1 minute)

To understand how **EDM works**, we first need to know the **KDD process**, which stands for **Knowledge Discovery in Databases**.

The KDD process consists of several steps.

First, **Data Cleaning**, where we remove noise and handle missing values.
Next, **Data Integration**, where data is combined into a single dataset from multiple sources.
Then, **Data Selection**, where we choose the relevant data needed for analysis.
After that, **Data Transformation**, where raw data is converted into suitable formats for mining.

Next comes **Data Mining**, where algorithms are applied to extract patterns and relationships from the data.
Then, **Pattern Evaluation**, where we identify which patterns are meaningful and useful.
Finally, **Knowledge Representation** Come, here we present our refined/processed data, which is called information or knowledge.

Slide 5: Common Techniques of EDM (1 minute)

Now, let's look at some **common techniques used in Educational Data Mining**.

First, **Classification** —
In classification, we predict a **predefined class**, for example, a student will **pass or fail,** based on factors like quiz scores or attendance.

Second, **Clustering** —
In clustering, we group **similar data points together** without any predefined classes.
For example, students can be grouped into **high, medium, and low performers** based on their academic performance.

Third, **Association Rules** —
Here we identify **relationships between two or more features**.
For example, students who study late at night aften submit assignments late.

Finally, **Prediction** —

Here we predict future outcomes based on historical data.
For example, predicting a student's **final grade** using their past academic performance.

These techniques help educators **understand student behavior more clearly** and make decisions effectively.

Slide 6: EDM vs Traditional Machine Learning (1 minute)

Let's discuss the **key differences between EDM and Traditional ML**.

Although Educational Data Mining uses machine learning techniques, there are some important differences between them.

Educational Data Mining mainly focuses on:

- Improving student learning outcomes
- Supporting teachers and educational institutions
- Early identification of at-risk students

On the other hand, **Traditional Machine Learning** focuses more on:

- Improving model accuracy and performance
- Automation and prediction
- Often complex or black-box models

In short, **EDM is education-centered**, while **traditional machine learning is technology-centered**.

Slide 7: Importance of EDM (50–60 seconds)

Educational Data Mining is important for several reasons.

First, it can **handle large educational data** generated from grades, attendance, logs, and assessments.
Second, it helps **bridge the information gap** by converting raw data into meaningful insights/information.
Third, EDM supports **early intervention** by identifying at-risk students before serious academic failure occurs.

Additionally, EDM helps educators **understand learning patterns**, improve **decision-making**, and **adapt to personalized learning needs** based on individual strengths and weaknesses of a student.

Slide 8: Applications of EDM (1 minute)

This slide shows the **real-life applications of Educational Data Mining**.

EDM is applied in :

1. Student Performance Analysis
2. Early Warning Systems
3. Personalized & Adaptive Learning
4. Curriculum & Teaching Improvement
5. Institutional Decision Making
6. Assessment & Feedback Optimization
7. Social Learning Analytics

In real educational settings, these applications are already producing strong results.

A research study published in **2024** shows that **student performance analysis** can predict final grades with an accuracy of **88.6 percent** using academic records.

Another research study shows that **EDM-based early warning systems** can **reduce dropout rates by 14 percent** by identifying at-risk students early.

In addition, A study published in **2025** shows that **automated essay scoring systems are very close to human grading**, with a **0.88 to 0.91 correlation**, and they provide **instant feedback**

These applications show how EDM is already transforming modern education.

Slide 9: Case Study – MIT Project-Based Learning System (1 minute)

Before using **Educational Data Mining**, MIT faced many problems –

First, the **dropout rate was very high**—about **38 percent of students** dropped out of advanced CS courses.

Second, **team problems were detected very late**. Most of the time, **teams were formed randomly**, so students had **unequal skills**, and some students did most of the work.

Also, many projects failed because students chose **overambitious and unrealistic project goals**, and also **limited visibility** into team collaboration during the coding process.

EDM Solution ***(continue on the same slide, calm pace)***

To solve these problems, MIT introduced an **Educational Data Mining–based system**.

The system analyzed **GitHub data**, such as how often students commit code, review code, and contribute to the project.

It also used an automatic project scoping advisor that suggested **realistic project goals using data from past projects**.

A **real-time team health dashboard** showed teamwork imbalance and technical problems early.

The system also analyzed **peer feedback and messages** to detect early signs of conflict inside teams.

As a result:

Project completion rates increased by **31%**

Team satisfaction improved by **42%**

Instructor intervention time reduced by **35%**

Participation of underrepresented students in leadership roles increased

This case clearly shows the **real-world impact of Educational Data Mining**.

Final Slide 10: Challenges & Conclusion (45 seconds)

Despite its advantages, EDM also faces some challenges.

These include:

Ensuring **data privacy and security**

Handling **missing, noisy, and inconsistent data**

Integrating data from multiple educational sources

Maintaining **interpretability** of complex models

Addressing **ethical issues and bias**

In conclusion,
Data Mining transforms raw data into meaningful knowledge, and **Educational Data Mining** applies this power to improve education.
EDM enables early identification of at-risk students, supports data-driven decisions, and promotes personalized learning.
Despite existing challenges, EDM has **strong potential to shape the future of education**.

Short Conclusion:

Educational Data Mining (EDM) uses data mining techniques to extract meaningful insights from educational data. Overall, EDM plays a key role in improving teaching quality, learning outcomes, and personalized education.

Closing
Thank you all for your kind attention.